Faithful on Prose, Unanchored on Reasoning: A Position-Domain Calibration of the Jacobian Lens

Manabu Higashida
D3 Center, The University of Osaka, Osaka, Japan
manabu.higashida.cmc@osaka-u.ac.jp ORCID: 0009-0004-7393-2957

Abstract

The Jacobian lens (J-lens) reads which tokens a language model is positioned to verbalize, and its released artifacts are being inherited as defaults. Those artifacts are fitted with a maximum sequence length of 128 tokens, a setting that appears in the distributed configuration files and not in the main text of the canonical write-up; readers are applying the lens thousands of tokens deep. We calibrate fidelity against position on OLMo-3-32B, with the decision table, the thresholds, and the axes of both figures committed before any measurement existed. On prose — the lens's home domain — fidelity does not decay with depth. On reasoning traces it does, but the finding does not survive contact with its own coordinate system: the same measurements read as a degradation of +0.82 dex against one baseline cell and an improvement of −0.14 dex against another. We report both coordinates and treat that dependence as a primary result rather than a nuisance. Extending the prompt axis from one prompt to three, we find the deep-prompt degradation reproduces in all three, and that in two of them the decision is carried by a deep-rank measure alone while head-of-distribution rank correlation stays flat — the failure mode we had named at design time in order to guard against it. We give the resulting problem a name: *the calibration was anchored neither outside nor inside the instrument* — no external oracle to check it against, and no internal origin to measure from. This is not a property of one lens. Anyone who inherits a disposition-reading instrument with its default settings inherits the same problem. Give us a ruler for the order of reasoning.

Keywords: interpretability; Jacobian lens; calibration; position domain; pre-registration; language models

1 Introduction

The Jacobian lens (J-lens) was introduced by Gurnee et al. as an instrument for identifying internal representations that are readily available for verbal report [1]. For each token in the vocabulary, "the Jacobian lens identifies a vector representation that encodes the potential for the model to verbalize that token in the future" [1]. Collectively these vectors are termed the *J-space*. Fitted lenses have been released publicly for several models, and downstream work — ours included — has begun to use them.

An instrument that ships with defaults will be used with those defaults. This paper asks a question that the instrument's own documentation does not answer: *over what range of token positions is the lens calibrated?*

The canonical write-up describes the fit as an expectation "over the source position t, all subsequent positions t' within the context, and a corpus of one thousand prompts sampled from a pretraining-like distribution" [1]. Its main text states neither the sequence length used in fitting nor any dependence of fidelity on token position within the context.[1][2] The actual setting — `max_seq_len` = 128 — appears only in the configuration files distributed alongside the fitted lenses, and it is the same across all six released lenses. A user therefore cannot learn the position domain from the paper; they learn it by opening a config file.

We do not present this as a defect of the original work: a research note may reasonably omit such a parameter. We present it as a fact with a consequence. The position domain is inherited undocumented, and the distance between where the lens was fitted (128 tokens) and where it is applied (we ourselves had been reading at 8192) is a factor of 64; this study measures out to 16384, a factor of 128.

This paper calibrates that gap and reports what calibration could and could not establish. Section 2 describes a procedure whose central feature is that the decision table and both figure frames were frozen and committed before any measurement existed. Section 3 reports the measurements. Section 4 states what we take to be the main contribution, which is not a number.

Our position is a calibration, not a rejection. The original authors are explicit that "the Jacobian lens is an imperfect tool, which we believe only approximately and incompletely captures the model's underlying workspace structure" [1]. This work is in the same direction as that sentence.

2 Method

2.1 Calibration target

Lens	J-lens, public artifact (md5 `c73a32d1...`)
Model	OLMo-3-1125-32B, bfloat16, no chat template
Read-out band	layers 26–56 (31 layers: 8 full-attention, 23 sliding)
Fit setting	`max_seq_len` = 128; `wikitext-103-raw-v1`
Prior use	our earlier runs applied it at `max_seq_len` = 8192

2.2 Quantities: calibrate on what the instrument consumes

A calibration is only as good as its choice of measured quantity. We report five.

KL	KL(softmax(model) ‖ softmax(lens)), band median
top-1	argmax agreement, band mean
Rank correlation	Spearman on the model's top-K ($K = 1000$), band median
Word-group rank deviation	$\lvert \log_{10} \mathrm{rank}_{\mathrm{lens}} - \log_{10} \mathrm{rank}_{\mathrm{model}} \rvert$, band median
Baseline contrast	the same, against a vanilla logit lens [3]

The fourth quantity requires comment, because it is the one that ultimately decided this paper. Rank correlation over the model's top-1000 measures fidelity at the *head* of the distribution. The downstream use we care about — reading which of two word groups is ranked higher — consumes ranks that lie two to five orders of magnitude deep. Head fidelity does not guarantee depth fidelity: a lens whose head is faithful and whose flank is distorted could pass a decision table built on rank correlation alone. We therefore added *word-group rank deviation* as a fifth quantity at design time,

[1] The write-up does note that readouts are noisy in early *layers*; that concerns depth in the network, not position within the sequence.

[2] We searched the published page on 2026-08-03 for `max_seq`, `max_seq_len`, "sequence length", "context length", "context window", "token limit", "prompt length", "truncat" and "window size"; none occurs. We were not able to retrieve the appendices, so this is a statement about the main text only. The search predicates, the date and this limitation are recorded with the release.

and made the decision the worst of {rank correlation, word-group rank deviation}, specifically to guard against that failure mode. Section 3.3 reports what happened.

2.3 The freezing procedure

The methodological commitment of this paper is procedural:

1. The decision table — thresholds, decision group, baseline, minimum cell sizes — was frozen and committed *before* measurement.
2. The figures' axes, bucket boundaries and decision lines were frozen and committed while no data point existed: the plotting scripts draw an empty frame if the results file is absent.
3. When results arrived, the table was applied mechanically. The operator moved neither a threshold nor a verdict.
4. Where data did not fit the frozen frame, the frame was not widened. Out-of-range points are marked with triangles and their values printed.

Each commit timestamp is verifiable evidence that the freeze preceded the measurement. The procedure fired twice during this study. Once, the frozen figure axis (an absolute Δ in dex) caught a unit mismatch in the prose of the decision table (a ratio with a floor); the frame was right and the text was corrected to match it. Once, data exceeded the frozen limits in both directions and was received with triangles rather than a wider axis — so that "it did not fit" survived as information.

2.4 Sample

Prose corpus (W)	4 concatenated `wikitext-103` [2] texts, $\geq$16,400 tokens each (the lens's fit dataset; article boundaries recorded in provenance)
Reasoning traces (G)	8 generation traces over 3 prompts: prompt 1 (458 tokens) — 2 traces prompt 2 (470 tokens) — 2 traces prompt 3 (501 tokens) — 4 traces
Positions	8 evenly spaced per frozen bucket; one forward pass reads all
Total	924 measurements (700 initial + 224 in the multi-prompt extension)

The three prompts are edit variants of one base task and therefore share prefixes: prompt 3 diverges from prompts 1 and 2 at token 86, and prompts 1 and 2 diverge from each other at token 354. Limitation 3 (§5) treats the consequences, which are substantial.

The sampling rule for the extension was itself frozen before the run: scanning seeds in ascending order, take the first two traces with more than 512 tokens. The length cutoff is not arbitrary but a requirement of the position grid — a shorter trace has its bucket ceilings truncated, so the same absolute positions can no longer be compared across prompts. One candidate trace was disqualified by this rule.

2.5 Reproducibility and positive controls

Two applications, identical settings	all values $\Delta = 0$; ranks 48/48 identical
Same, with truncation active	all values $\Delta = 0$; ranks 47/47 identical
Implementation difference at equal length	word-group rank deviation, median 0.0109 dex
Control A (aggregator)	reproduces the published re-aggregation: 170 values, 0 differing
Control B (measurement)	reproduces a certified trace: 520 values, 0 differing

Read-out is deterministic, and the implementation difference is below half the decision threshold. For the multi-prompt extension we did not assume that the extension's instrument was a faithful copy of the frozen one; we measured it, with Controls A and B, before taking any new measurement. A mismatch in either was a pre-registered stopping condition that would have halted the run before it began.

One caveat belongs here. Prompt-side read-outs agree *exactly* only when the run configuration — the set of positions read together, and the sequence length — also agrees. Across pairs differing only in run configuration, with prompt and prefix identical, we observed a median of 0.009 dex and a maximum of 0.031 dex, one tenth of the decision threshold ($n = 15$ positions, single prompt pair). We do not identify a mechanism.

3 Results

3.1 Findings independent of the coordinate system

Some of what we measured does not depend on how positions are indexed. We state those first, and we state *in what sense* each is independent, because the senses differ.

Finding	Value	Kind of independence
Prose improves across all buckets (eager)	rank corr. −36.4 to −54.5%	**Structural**: prose has no prompt/generation split, so it is not even a subject of the transform
Rank-correlation degradation agrees across coordinates	+11.6/+11.6; +18.3/+13.2	**Empirical**: it happened here; not a theorem
Deepest bucket is out of domain	+0.754 dex $\wedge$ top-1 −6.5 pt	**Procedural**: two requirements, settled inside one attention implementation
Fit `max_seq_len` shifts every reading uniformly	∼0.2 dex, flat in position	Empirical — a first-order warning for practitioners
J-lens beats vanilla logit lens	+0.063 to +0.195	Empirical (implementation sanity)

On prose, the lens is at home — in one universe. The scope matters here, and dropping it would put the claim in conflict with our own ledger. Under `eager` attention (buckets B1–B5, `max_seq` 8192), the prose corpus improves on *both* measures at every bucket: rank correlation by −36.4 to −54.5%, and word-group rank deviation by −0.367 to −0.175 dex. Under `sdpa` (B1–B6, `max_seq` 16384), the deep buckets *degrade*: word-group rank deviation reaches +0.437 dex at B5 and +0.380 dex at B6, crossing the 0.30 line, and rank correlation degrades +22.6% at B6.

The first half of our title therefore holds *in the eager universe*. We cannot write, without qualification, that prose improves at every bucket. Note also that the two runs vary attention implementation and `max_seq_len` together, by the frozen attention plan; the prose gap between them is therefore not attributable to position alone. The same confound is quantified below as 0.2089 versus 0.0109 dex. We do not identify a mechanism.

Nor should this be read as "position is fine." The prose corpus is the lens's own fit dataset (Limitation 4), which favors it. What the finding supports is narrower: *in the eager universe, on prose, position itself does not destroy fidelity.*

A configuration parameter moves every reading at once. Comparing implementations at *equal* sequence length gives a median word-group rank deviation of 0.0109 dex, inside the frozen guard of 0.02. The corresponding term in the main run was 0.2089 dex. The difference — a factor of nineteen — is attributable to `max_seq_len`, not to the attention implementation. A user who inherits the default inherits a uniform shift of roughly 0.2 dex in every reading, without any signal that they have done so.

3.2 Both coordinates, and why the dependence is the result

Figure 2 is a re-aggregation of measurements contained in Figure 1; **no new model run was performed**. The juxtaposition holds for the 352 reasoning-trace measurements (the 348 prose measurements have no prompt/generation split and remain in absolute coordinates, so they do not appear in Figure 2). What differs between the two figures is the choice of baseline cell, and nothing else.

	B4 / O4	B5 / O5
Absolute coordinates (baseline B1, start of prompt)	+0.618 dex	+0.820 dex
Relative coordinates (baseline O1, first generated tokens)	+0.063 dex	−0.144 dex

The same points read as a degradation of +0.82 *dex or an improvement of* −0.14 *dex depending on where the origin is placed.*

There is consequently no option to report only one coordinate: a single-coordinate report is refuted by one re-analysis. We do not decide which coordinate is correct. That we cannot decide is itself the result.

Only one quantity is coordinate-dependent — the word-group rank deviation; rank-correlation degradation agrees across coordinates at B4/O4 and stays close at B5/O5. Its shape is not a gradual decay with position but a step at the prompt/generation boundary (B1 0.653 → O1 1.425, a step of +0.77 dex) followed by near-flatness within the generated region (O4 +0.06, O5 −0.14). Comparisons *within* the generated region are thus made under nearly common conditions — while the generated region as a whole floats free of any anchor.

The relative buckets, and this figure's empty frame, were committed before the aggregation was computed. Re-slicing coordinates is the easiest operation to perform after seeing the data, which is exactly why it was frozen first.

3.3 Depth within the prompt, across three prompts

Within the prompt, the deepest bucket degrades. The initial measurement saw this on a single prompt read four times, which licensed nothing about prompts in general. We added two prompts (eight traces over three prompts) and froze the reading table before the run: with the shallowest bucket P1 as baseline, "degraded" means a rank-correlation drop $\geq$ 30% *or* a word-group rank deviation $\geq$ +0.30 dex — the thresholds of Figure 1, applied mechanically.

		Rank correlation		Word-group rank deviation	
Prompt	Length	P2	**P3**	P2	**P3**
1	458	−72.1%	+**0.4**%	+2.000	+**1.152**
2	470	−72.1%	+**0.4**%	+2.000	+**0.530**
3	501	−12.0%	+**51.6**%	+0.389	+**0.812**

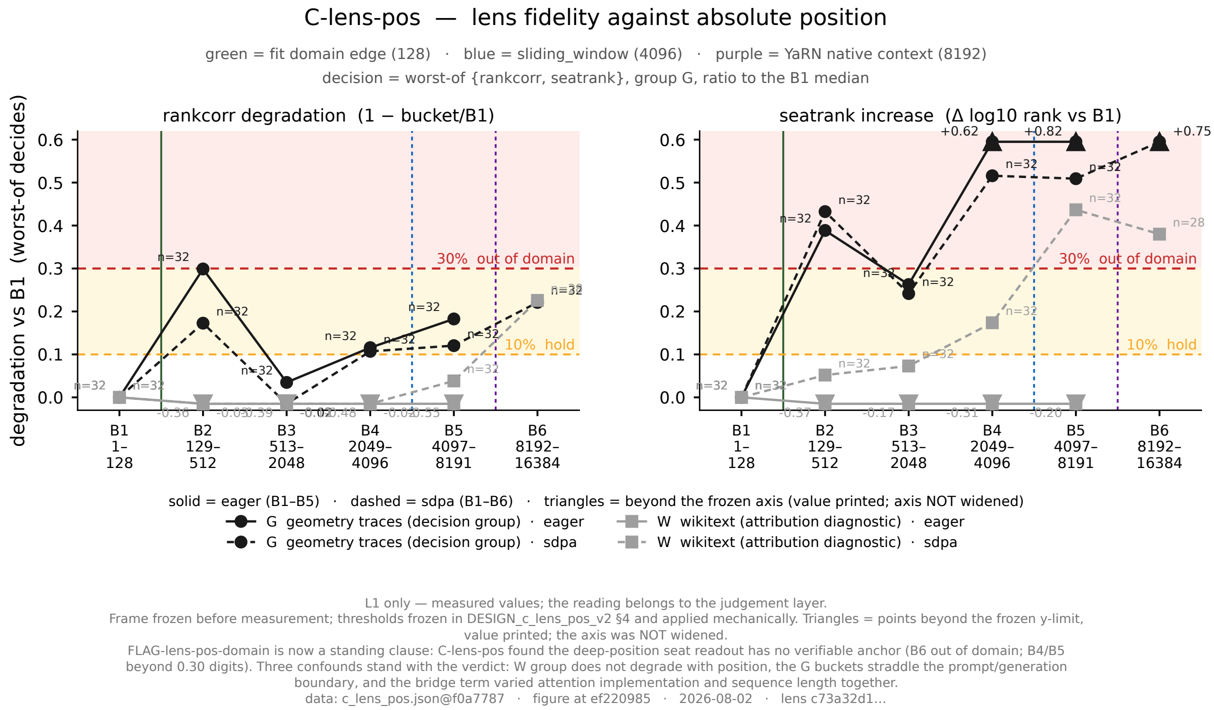

Figure 1: Lens fidelity against absolute position. Degradation of two fidelity measures relative to the first bucket (B1, tokens 1–128), for the reasoning-trace corpus (G, decision group) and the prose corpus (W, attribution diagnostic). Left: head-of-distribution rank correlation, as a ratio to the B1 median. Right: word-group rank deviation, as an absolute difference in $\log_{10}$ rank. The decision is the worst of the two, on group G only. Dashed horizontal lines are the pre-registered thresholds; vertical guides mark the fit-domain edge (128), the sliding window (4096) and the native context length (8192). The axes, buckets and decision lines were committed before any measurement existed. Triangles mark points beyond the frozen limit with the value printed: *the axis was not widened.*

Prompts 1 and 2 share their first 354 tokens; their P1 baselines and P2 entries are therefore identical by construction, not independent replicates (see Limitation 3). The repeated "−72.1% / +2.000" across the two rows is the visible symptom of this. The pre-registered comparison — P3 against P1 — is unaffected: P3 spans the divergence point at token 354, so all three rows differ there.

The pre-registered comparison is P3 versus P1 (bold). The P2 column is reported as an observation *outside* the pre-registered comparison: these are measurements on the same frozen grid, and withholding a non-monotone shape would be selective reporting.

All three prompts fall in the degraded branch. The deep-prompt degradation is therefore not specific to a single prompt.

Which measure carried the decision. For prompts 1 and 2, rank correlation is flat at +0.4% — it does not approach the 30% threshold — while word-group rank deviation crosses the 0.30 dex threshold by factors of 1.8 to 3.8. For prompt 3 both measures fire. The scope of "alone" is therefore two of the three prompts, across both attention implementations (prompt 3 fires on both measures: +51.6% eager, +42.8% sdpa).

Had we decided on rank correlation alone, two of the three prompts would have read as undegraded.

The measure that carried the decision is the one added at design time against a named failure mode: *a lens whose head is faithful and whose flank is distorted could pass a decision table* (§2).

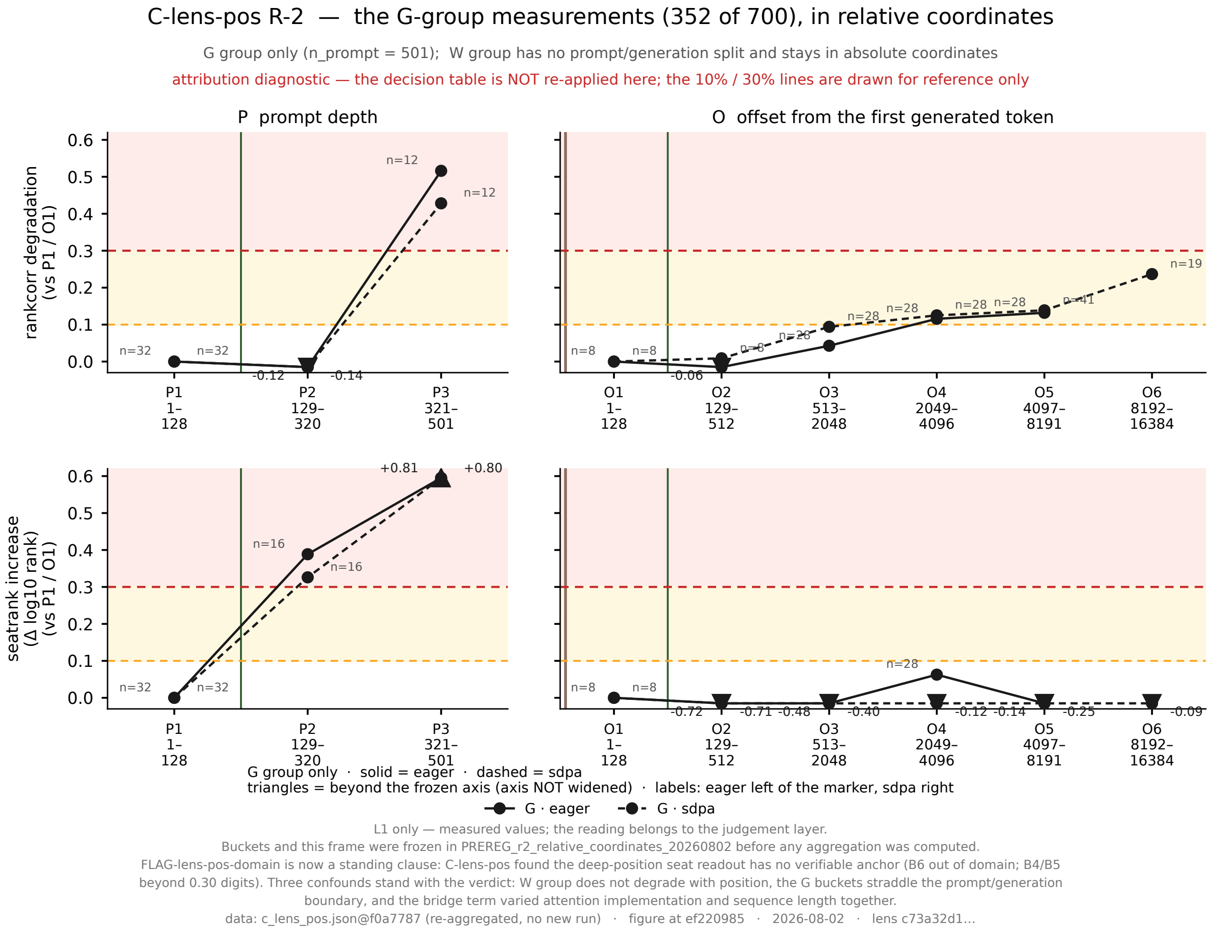

Figure 2: The same measurements in relative coordinates. The reasoning-trace measurements of Figure 1 (352 of the 700), re-bucketed by depth within the prompt (left column) and by offset from the first generated token (right column). **No new model run was performed.** The prose corpus has no prompt/generation split and stays in absolute coordinates, so it does not appear here. Baselines are the first bucket of each coordinate system, not B1. The 10%/30% lines are drawn for reference only; the decision table is not re-applied. Buckets and this frame were committed before the aggregation was computed.

That failure mode appeared in measurement. We report this as an occurrence of a hazard we had named, not as a prediction confirmed — it was a design rationale, not a registered prediction with a stake attached.

Shape. In two of the three prompts the largest deviation lies in the *middle* of the prompt (P2) and shrinks toward the end. We therefore do not use any localization slogan about degradation living at the prompt's tail; that shape belongs to one prompt. No mechanism is claimed.

4 The contribution: a calibration with no anchor

4.1 No external anchor

An instrument that reads dispositions has no external oracle.

The lens is designed to read the *potential to verbalize a token in the future*. The only quantity available to check it against is the model's final-layer logits — the prediction of the *next* token. These two are different by construction. Agreement can therefore be measured; disagreement cannot be converted into a verdict of failure. A divergence may be (i) unfaithful extrapolation by the lens or (ii) a widening — correct, and expected — gap between disposition and next token. This study cannot separate them, and the obstruction is not one of sample size or compute (Limitation 5).

4.2 No internal origin

Nor is there an origin inside.

First, as §3.2 shows, the choice of baseline cell reverses the sign of "degradation" (+0.82 against −0.14 dex). Second, as §3.3 shows, agreement at the head of the distribution does not imply fidelity at the depth the instrument actually reads: cells exist in which head correlation is flat while deep ranks alone move the verdict. Where to place the origin, and at what depth to look, are not settled from inside the calibration.

4.3 The statement

> *The calibration was anchored neither outside nor inside the instrument.*

— no external oracle to check it against, and no internal origin to measure from.

This structure is not specific to this lens. Anyone who inherits a disposition-reading instrument together with its default settings inherits the same problem. The contribution of this paper is not a number; it is a name for that problem.

We do not claim the lens is unusable. On prose it is faithful (§3.1). The claim is narrower and, we think, more useful: *readings taken deep inside a reasoning trace have no verifiable anchor.*

> *Give us a ruler for the order of reasoning.*

5 Limitations

1. Single model (OLMo-3-1125-32B), single stack (PyTorch, Apple MPS, bfloat16).
2. Single lens (md5 `c73a32d1...`), single band (layers 26–56).
3. **The reasoning-trace corpus is 3 prompts over 8 traces, but these are not independent samples, for two reasons.** (a) In a causal language model the state at a prompt position depends only on tokens up to that position, so multiple traces of one prompt reproduce identical values on the prompt side. (b) The three prompts are edit variants sharing prefixes (divergence at tokens 86 and 354), so before a divergence point, different prompts return *exactly* the same value — we verified this: two pairs of prompts, 13 positions each, gave $|\Delta| = 0.0000$ in word-group rank deviation. Consequently the correct unit for counting independent values is (prefix, position), not (prompt, position) and not the number of rows. Across the three prompt-internal buckets the counts are, respectively, rows 64/32/24; $(\text{prompt}, \text{position})$ 24/12/9; $(\text{prefix}, \text{position})$ **11/8/8**. We therefore do *not* claim to have doubled n. The claim is only that positional generalization widened from one prefix to three. *Disclosure:* the pre-registration froze the distinct count under $(\text{prompt}, \text{position})$, giving an expected 9; the sharper predicate $(\text{prefix}, \text{position})$, giving 8, was identified after the run. Prefix sharing is a property of causal language models, not a defect of the sample.
4. The prose corpus is the lens's own fit dataset, which favors it.

5. **Fidelity is checked against the model's final-layer logits, i.e. the next-token prediction, while the lens is designed to read near-future disposition.** The two differ by construction. A divergence cannot be attributed to (i) unfaithful extrapolation rather than (ii) a growing gap between disposition and next token. This is not a shortfall of samples or compute: *an instrument that reads disposition has no external anchor for correctness.*

6. Position buckets were frozen in absolute coordinates, and in reasoning traces they straddle the prompt/generation boundary; the re-aggregation in relative coordinates was frozen and committed before it was computed.

7. Readings depend slightly on run configuration: "same prompt, same value" does not hold in general — agreement requires the run configuration to match as well. The magnitude is given in §2 and is far below the decision threshold; the decision is unaffected. Practitioners must fix not only the configuration of the model but the configuration of the reading.

8. Traces for the multi-prompt extension were selected by a length rule (> 512 tokens); the distribution over generation-side offsets is therefore not an unbiased sample, and we do not read it.

6 Research workflow and AI involvement

This calibration was carried out through a two-layer workflow of large language models [5], described here because parts of it are load-bearing for the paper's warrant. An *execution layer* (agentic coding interface) implemented the harness, ran the measurements, and reported them in frozen documents. A separate *judgment layer* (chat interface) performed adversarial review of every design and result document under a single-identified-fatal-flaw discipline, and froze the pre-registrations and reading tables. The two layers are separated by a deliberate information asymmetry: the judgment layer has no access to the codebase, the raw outputs, or the subject model, and reads only the frozen documents.

Three boundaries fix responsibility. First, the subject of measurement is OLMo-3-32B and the released lens; no workflow model is a measurement subject. Second, no language model adjudicates an outcome: every quantitative verdict is taken against a threshold frozen before its measurement and applied mechanically. Third, every load-bearing decision — thresholds, terminology, framing, and the final wording of the claim — was reserved to and made by the human author, who reviewed all frozen documents and bears sole responsibility for every claim.

The workflow's own errors are reported rather than removed. Two are material to this paper. The scope of the prose finding (§3.1) was dropped by the execution layer in a draft and restored in review after collision with the ledger. The sharper counting predicate in Limitation 3 was identified only after the run, and is disclosed as such rather than presented as pre-registered.

Data and code availability

All measurements are released with per-position values retained: the main run (700 measurements), the control re-measurement, the relative-coordinate re-aggregation (no new run), and the multi-prompt extension (224 measurements), together with the determinism and neutrality checks. The measurement code imports the lens, the word groups, the rank computation and the token mask verbatim from the upstream implementation. The frozen documents — the decision table and the two pre-registrations — are released with their commit timestamps, which are the evidence that the freeze preceded the measurement — as post-hoc English renderings, with the Japanese originals fixed by `sha256` (see `PROVENANCE.md` in the release). Figure scripts are included and produce an empty frame when no data file is present.

The package is available at `https://github.com/moyhig-ecs/FaithfulProse`. Every file is listed in `MANIFEST.json` with its `sha256` both as it ran and as released, together with an enumeration of every substitution made for release.

Acknowledgment

This work was supported by JSPS KAKENHI Grant Number JP26K06399. Manuscript text was drafted within the two-layer workflow described in §6 [5]; all claims, thresholds and final wording are the author's.

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